

Adriel Niyodusaba

Software Engineer (AI & Backend) - engineeradriel.com

Seoul, South Korea · Open to US / Canada / Korea / Remote

SUMMARY

Software Engineer specializing in **AI-powered backend systems and applied machine learning**, with 4+ years of experience building and scaling **production products** in media, video, and language technology. Core builder of **Slid, Summary, and Dipclip**, shipping end-to-end systems spanning **AI pipelines, backend APIs, frontend tools, and infrastructure**. Strong background in **RAG systems, speech & video AI, agent pipelines, and cloud-native deployment**, with growing focus on **security, Web3, and AI safety**.

CORE TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Go, Rust
 - **Backend:** Django, FastAPI, Node.js, Nest Js, Gin
 - **AI / ML (Applied):** Langchain, Langgraph, Autogen, TTS, STT, LipSync, RAG, LLMs
 - **Infra:** AWS, GCP, Docker, Kubernetes, Terraform
 - **Datastores:** PostgreSQL, MySQL, Redis, Pinecone, MongoDB
 - **Web3:** Wallet Development
 - **Media:** ffmpeg, audio/video processing pipelines
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PROFESSIONAL EXPERIENCE

Software Engineer — AI & Backend Bebridge · Seoul / Global Sep 2022 – Present

Software Engineer (Intern → Full-time) Dec 2020 – Aug 2022

Lead engineer on **SaaS products: Slid, Summary, and Dipclip**, shipping production AI systems end to end, later expanding into **B2B media AI, client projects**, and **Web3 initiatives**.

PRODUCT EXPERIENCE

SLID — AI-POWERED NOTE-TAKING PLATFORM FOR ONLINE LEARNING

Lead Software Engineer

Seoul, South Korea | Dec 2020 – Present

- Joined as one of the **first two engineers**; evolved from frontend intern into **lead engineer owning core product systems**.
- **Refactored the Chrome extension frontend** and **designed, built, and shipped** a cross-platform **desktop app (macOS & Windows)** from scratch, handling code signing and releases; desktop app grew to **~50% of active users**.
- Owned the **entire backend and cloud infrastructure** (APIs, databases, background jobs, deployments); built **global infrastructure** that significantly reduced international latency and enabled worldwide usage.

- **Designed and implemented recurring subscription payments**, enabling sustainable monetization and long-term growth.
- Introduced **AI transcription (Whisper)** during an internal hackathon, which became the foundation of Slid's **Auto-Note** feature.
- Represented Slid at **CES 2023 (Las Vegas)**, presenting the product and its AI capabilities to an international audience.

Tech: TypeScript, React, Electron, Python, Node.js, FastAPI, PostgreSQL, ffmpeg, Whisper, AWS, Docker

SUMMARY — AI-POWERED VIDEO DIGEST MOBILE APP

Lead Backend Engineer

Seoul, South Korea

- Built and owned the **backend and AI pipeline** for a B2C mobile app that delivers **automatic summaries of newly uploaded YouTube videos**.
- Designed a **channel pooling and ingestion system** to monitor thousands of subscribed channels and trigger near-real-time processing on new uploads.
- Implemented end-to-end **AI pipelines**: video ingestion → transcription → LLM-based summarization → storage → user notification.
- Architected scalable backend services and databases to support **~10,000 users** with reliable, low-latency delivery.
- Optimized processing workflows to balance **cost, speed, and quality** in production AI workloads.
- Contributed to product strategy and system design decisions as the platform scaled.
- Represented Summary at **CES 2023 (Las Vegas)**, presenting the product and its AI capabilities to an international audience.

Tech: TypeScript, NestJS, React Native, Python, FastAPI, MongoDB, PostgreSQL, Whisper, GPT-4o, AWS Lambda

DIPCLIP — AI SHORT-FORM VIDEO GENERATION PLATFORM

Lead Backend / AI Engineer

Seoul, South Korea

- Built the **core AI pipeline** for converting long-form videos into **short, reel-style video segments** optimized for mobile consumption.
- Designed algorithms to **analyze video content**, identify key moments, and generate precise **time-range annotations** representing meaningful clips.
- Implemented backend systems to enable **seamless playback** that dynamically jumps across key segments while preserving narrative flow.
- Led system design and backend development for a **B2C mobile app** serving approximately **5,000 users**.
- Focused on scalable, production-ready AI pipelines balancing **latency, cost, and output quality**.
- Collaborated closely with frontend and product teams to align AI outputs with real user interaction patterns.

Tech: TypeScript, React Native, Python, FastAPI, PostgreSQL, Whisper, GPT-4o, AWS Lambda

B2B MEDIA AI & DUBBING SYSTEMS

Proteus — AI + Human Dubbing Agent

- Designed and built **Proteus**, an AI agent for multilingual video dubbing.
 - End-to-end pipeline:
 - Video ingestion and audio extraction
 - Transcription and translation via LLM agents
 - Web-assisted validation for translation accuracy
 - Voice cloning and translated audio generation
 - Iterative alignment to match **original audio duration** (non-trivial constraint)
 - Reduced **human dubbing time by ~90%**, with editors focusing on validation instead of creation.
 - System became a **core revenue driver**, generating **over half of company revenue** and securing **10+ B2B contracts**, including long-term clients.
 - Tech stack: ffmpeg, ElevenLabs, OpenAI, Gemini, LangGraph, custom orchestration.
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AI-Powered Dubbing Studio (Frontend + Backend)

- Contributed to a **web-based dubbing studio** used by human editors.
 - Integrated outputs from Proteus to allow:
 - Seamless playback and review
 - Rapid corrections and approvals
 - Finalization of high-quality dubbed content
 - Demonstrates ability to bridge **AI systems with human-in-the-loop UX**.
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SELECTED CLIENT PROJECTS

KOHI — Course Knowledge Graph & Semantic Search

- Built a **multi-level graph visualization** for thousands of courses.
 - Processed large-scale course data and embedded it into **Pinecone**.
 - Enabled **semantic search** alongside graph navigation.
 - Combined **data engineering, AI search, and UX-driven system design**.
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3Pro Investor Advisor — Financial AI Chatbot

- Built a chatbot answering investor questions based on **YouTube interviews with Wall Street experts**.
 - Implemented automated ingestion:
 - Detect new videos
 - Transcribe and embed content
 - Store in vector DB
 - Designed retrieval to prioritize **fresh content**.
 - Responses include **source video links and timestamps**, ensuring transparency and trust.
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Korean TTS Model Research & Deployment (Likelion) (Nov 2025)

- Researched and deployed multiple **open-source Korean TTS models** on GPU servers.
 - Benchmarked quality and latency across models including GPT-SoVITS, XTTS-v2, MeloTTS, Fish Audio, and others.
 - Helped client select and deploy a **production-ready TTS server**.
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Korean Lip-Sync AI Systems (Dec 2025)

- Researched, deployed, and optimized **MuseTalk** and **LatentSync** for Korean lip-sync.
 - Tuned models for client-specific performance and quality needs.
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WEB3 & SECURITY (RECENT)

Crypto Wallet & DeFi Risk Monitoring (Jan 2026 – Present)

- Designing **wallet security mechanisms** and **gas fee optimization**.
 - Built a system to monitor **Aave and Compound health metrics in real time**.
 - Authored a **fund-rescue smart contract** to protect user funds during abnormal conditions (testing phase).
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OPEN SOURCE CONTRIBUTION

- **Tenuo** — AI agent security framework enforcing **task-scoped authority**
<https://tenuo.dev/>
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EDUCATION & CERTIFICATIONS

- **BSc Bioengineering** — Yonsei University (Global Top #50, QS 2026) (2017-2022)
 - **AWS Solutions Architect – Associate**
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PROFESSIONAL EXPOSURE

- **CES 2023 Presenter (Las Vegas)**
 - **Silicon Valley Immersion (Dec 2025)** — 1 month meeting engineers, founders, and AI product teams
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AWARDS

National Science Grand Prize (Physics) · Korean Government Scholarship · Yonsei Academic Excellence · Korean Speech Contest Grand Prize · Slid Hero Award (Transparency) · Bebridge Hero Award (Focusing on Real Value)

LANGUAGES SPOKEN

English (Fluent) · Kinyarwanda (Native) · Korean (Advanced) · French (Intermediate)